

Project Documentation

A Comprehensive Measure of Well-Being - HDI Prediction System

1. Project Title

A Comprehensive Measure of Well-Being : Human Development Index (HDI) Prediction System

2. Abstract

Ee project lo Human Development Index (HDI) ni machine learning tho predict cheyadam kosam oka full web application build chesamu. 2021 global HDI dataset (195 countries) use chesi, Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, GNI per Capita ane 4 key indicators meeda Linear Regression model train chesamu (R-squared approx 0.98). Model ni Flask backend tho connect chesi, dark-themed web UI dwara users tama country details enter chesi HDI value predict cheyagalaru. Full project GitHub meeda host chesi, Render.com meeda deploy chesamu.

3. Problem Statement

Human Development Index ni official ga calculate cheyadam complex formula tho, multiple data sources tho jarugutundi, so normal users ki easy ga access cheyadam kastam. Prathi orka students, researchers, policy analysts ki oka simple tool levu, akkada nalugu basic indicators pettaga direct ga HDI estimate cheppedi. Ee gap fill cheyadam kosam, manam oka lightweight ML-powered web app build chesamu, ekkada user 4 values enter chesthe akkade real-time lo predicted HDI category (Low / Medium / High / Very High) chupistundi.

4. Features

- 4 simple input fields tho (Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, GNI per Capita) HDI predict cheyyachu.
- 2021 real global dataset (195 countries) meeda trained Linear Regression model, R-squared approx 0.98.
- Predicted HDI value tho patu category (Low / Medium / High / Very High) kuda chupistundi.
- Dark-themed, clean, responsive web interface.
- Flask backend tho pickle (.pkl) model serialize chesi fast prediction.
- Render.com meeda live deployment, ekkadi nunchi aina access cheyachu.

5. Tech Stack

- Programming Language: Python
- Data Handling: Pandas, NumPy
- Visualization: Matplotlib, Seaborn
- Machine Learning: Scikit-learn (Linear Regression, LabelEncoder, train_test_split, r2_score)

- Model Serialization: Pickle
- Backend Framework: Flask
- Frontend: HTML, CSS (dark theme)
- Development Tools: VS Code, Jupyter Notebook
- Deployment: Render.com
- Version Control: Git and GitHub

6. Installation Steps

- GitHub repository ni clone cheyandi: github.com/farzana1234-csd/HDI-Prediction-System
- Project folder loki navigate ayyi, requirements.txt lo unna packages ni install cheyandi (pip install -r requirements.txt).
- Dataset (hdi.csv) 'Dataset' folder lo undelaa confirm cheyandi.
- Terminal lo 'python app.py' run cheyandi - Flask development server start avtundi.
- Browser lo <http://127.0.0.1:5000> open cheyandi - HDI Prediction home page vastundi.

7. Usage Instructions

- Home page meeda 4 input fields kanipistayi - Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, GNI per Capita.
- Prathi field lo correct numeric value enter cheyandi.
- 'Predict' button click cheyandi.
- Next page lo Predicted HDI value display avtundi, together tho category (Very High / High / Medium) kuda chupistundi.
- 'Go Back' link tho malli home page ki return avvachu, mari values tho retest cheyachu.

8. Future Scope

- R-squared accuracy improve cheyadaniki Random Forest / XGBoost lanti advanced models try cheyachu.
- More socio-economic indicators (health index, environment index, gender equality) add cheyachu.
- Country-wise comparison charts and historical HDI trend graphs add cheyachu.
- Multi-language support (Telugu, Hindi) add cheyachu, so more users access cheyagalaru.
- User accounts and prediction history save cheyadam kosam database (MySQL / Firebase) integrate cheyachu.
- Mobile-friendly / PWA version build cheyachu.

9. Team Member Details

SmartBridge AI/ML Internship Project Team

- Shaik Farzana - Team Lead
- Chinthoti Raghavendra - Team Member
- Ahmad Kabir Shaik - Team Member
- Naga Sireesha Naralasetty - Team Member
- Sitha Ramaraju Velisala - Team Member

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